

20260827 Resolved Star Formation Population through RPS

Here we define a parameter call star forming ("survival") fraction (f_{SF}), which represents the fraction of star forming spaxel per Σ_* or R_e bin.

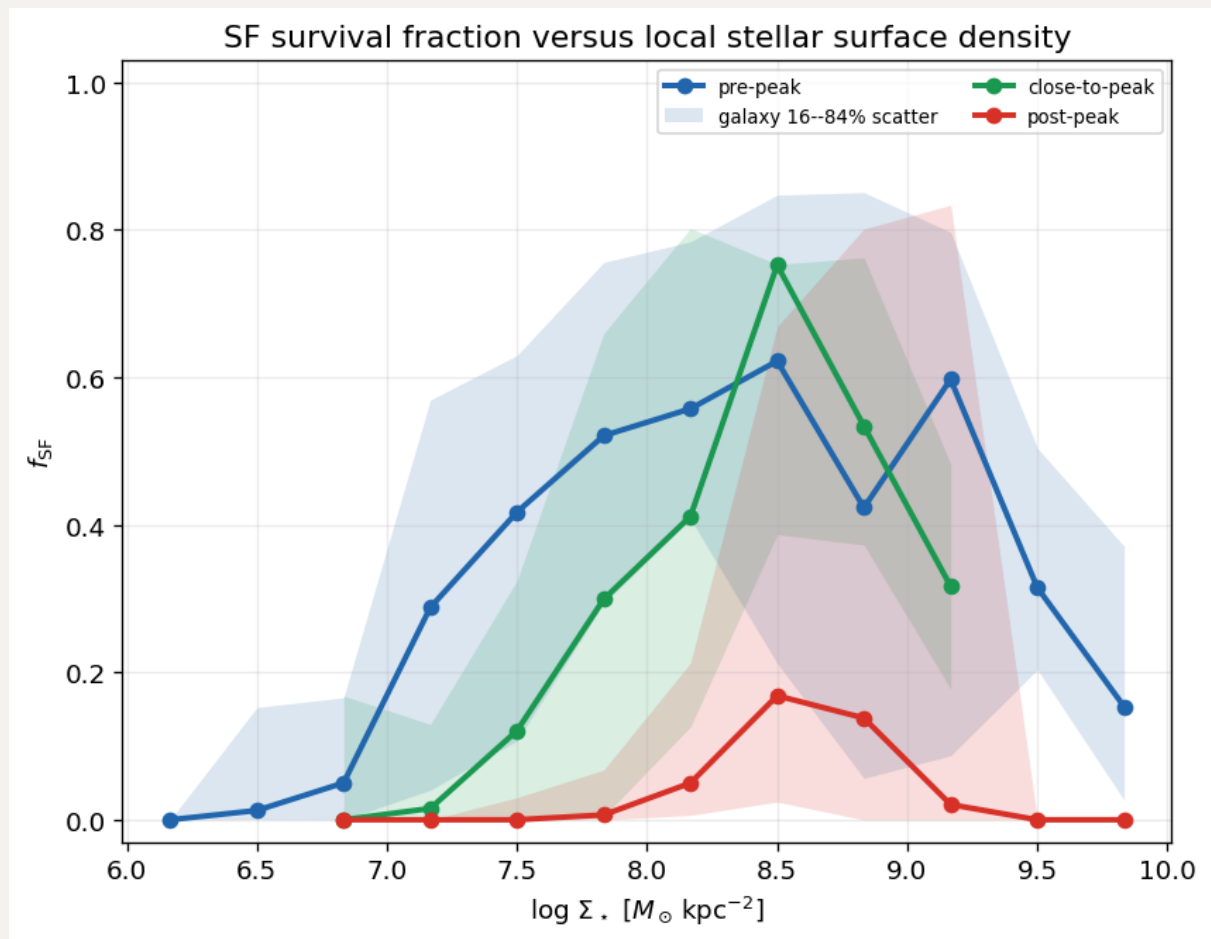
As usual, we define the star forming spaxels by HII BPT classification, $\text{EW}(\text{H}\alpha) > 6 \text{ \AA}$, and intrinsic $\sigma(\text{H}\alpha) < 45 \text{ km s}^{-1}$.

Also, we only select the galaxies that inclination less than 80° , so now we have 26: {'pre-peak': 8, 'close-to-peak': 5, 'post-peak': 13}

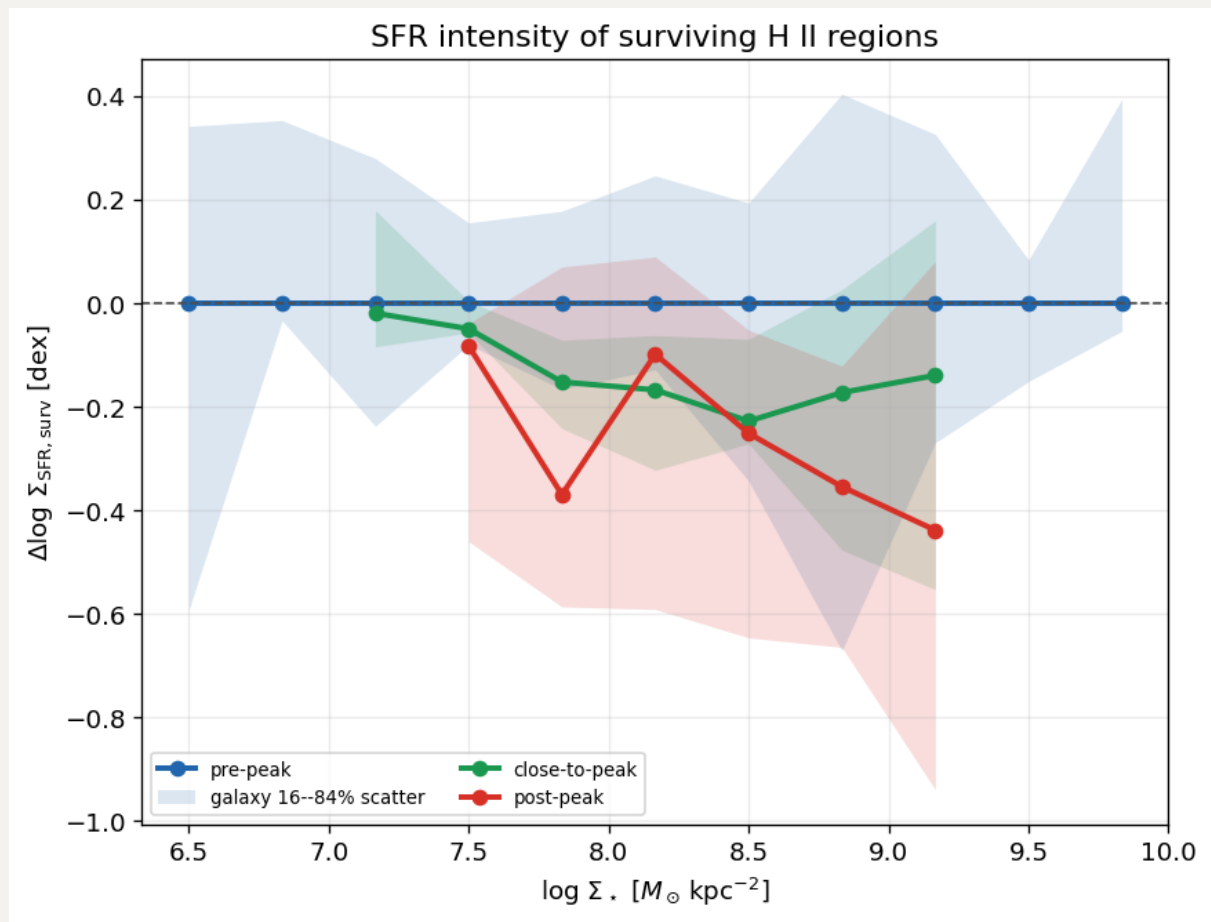
In general, the most promising change after peak ram pressure is a reduction in the fraction of usable regions that remain star forming, also the surviving star forming regions tend to be less star forming.

1. f_{SF} - Σ_* trend

First we look at SF survival fraction versus local stellar surface density. The pre-peak and close-to-peak medians increase toward intermediate-to-high Σ_* , while the post-peak median remains much lower across in all supported bins, though there are large scatters due to galaxy-to-galaxy diversity.

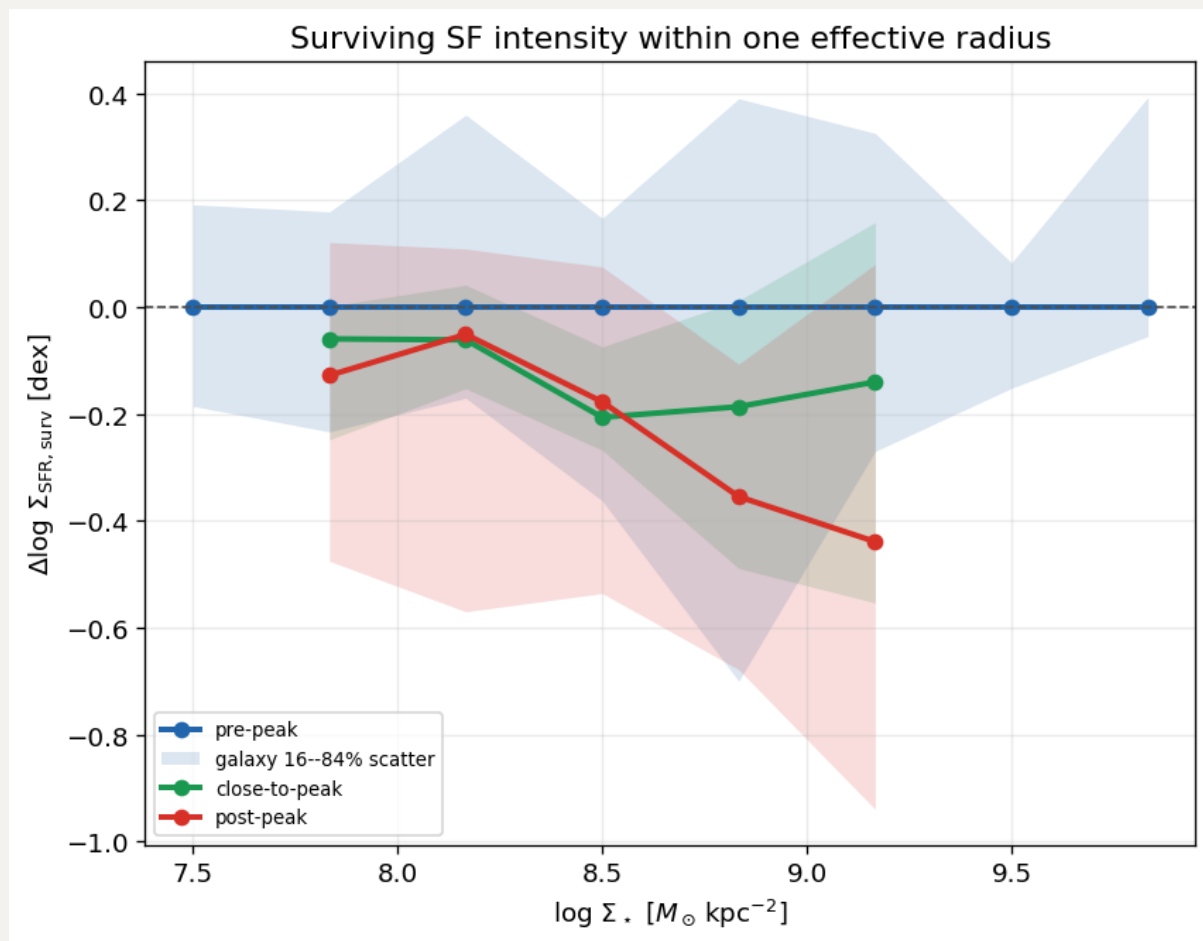
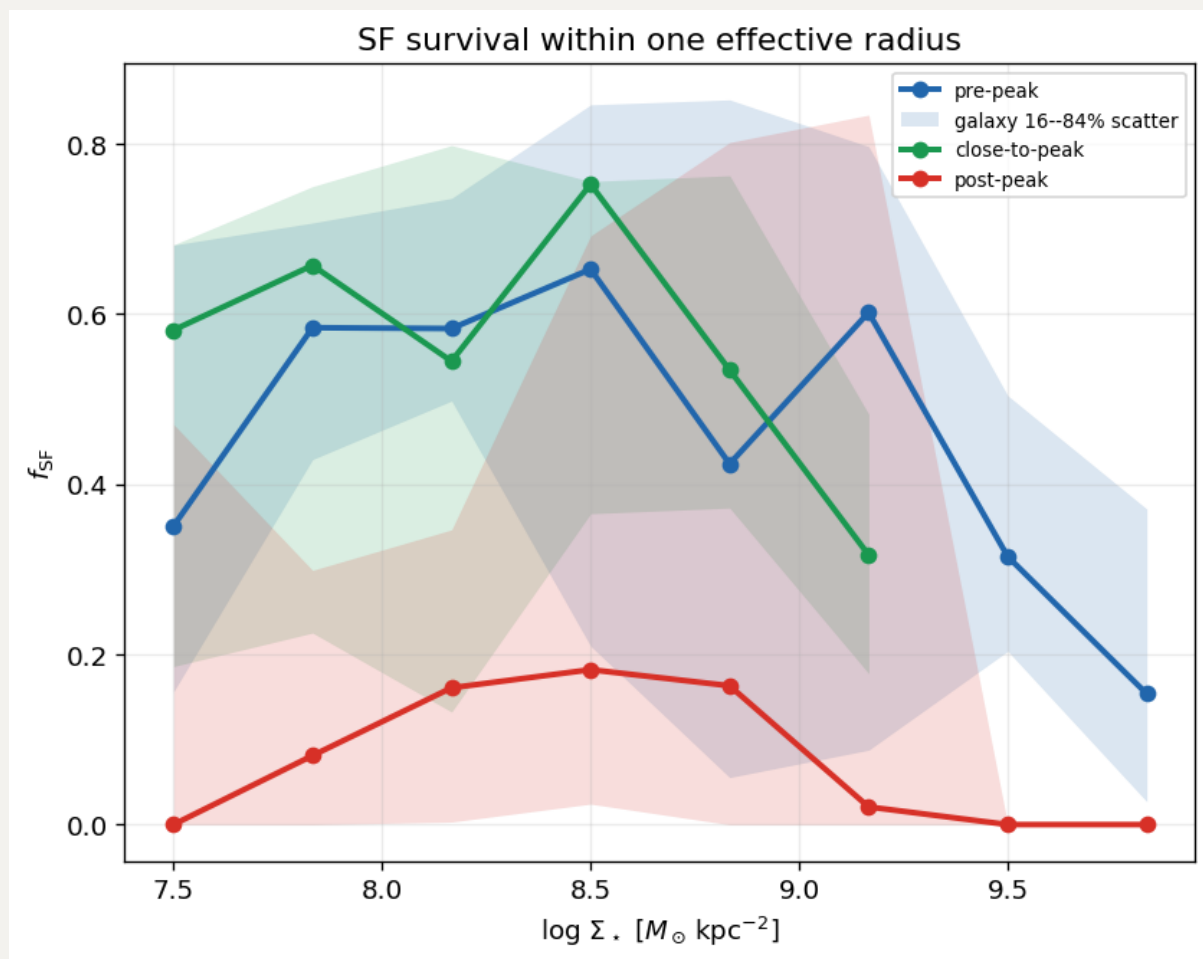


Then we plot the change of SFR on surviving H II regions, taking the median trend of SFR in pre-peak sample as the reference. Close-to-peak and post-peak medians are generally below the pre-peak reference at intermediate and high Σ_* .



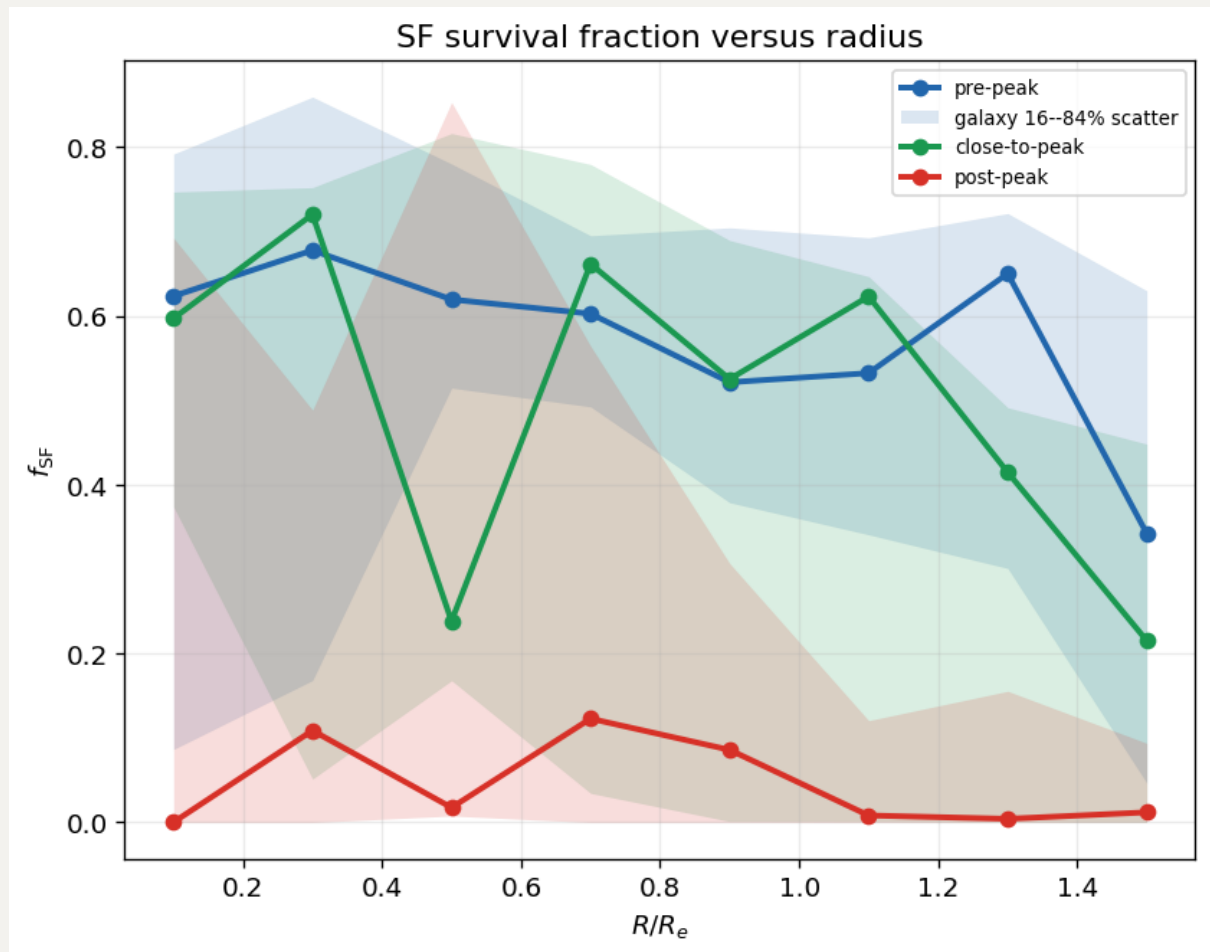
2. $f_{\text{SF}} - \Sigma_*$ trend within R_e

Restricting to SF spaxels within R_e shows similar results.

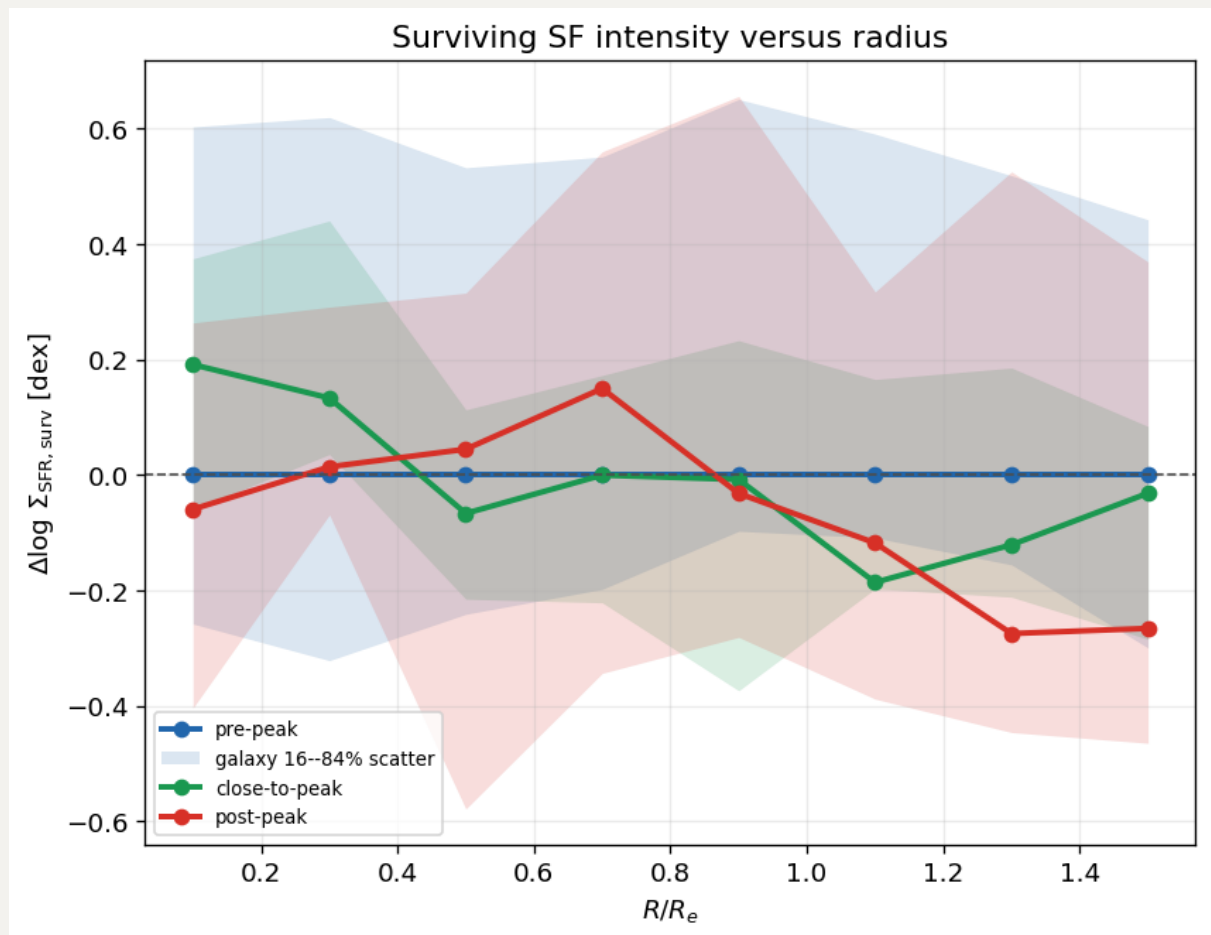


3. $f_{\text{SF}} - R_e$ trend

Now we look at SF survival fraction versus radius. Again, post-peak galaxies have very low median f_{SF} at all available radius, including inside R_e .

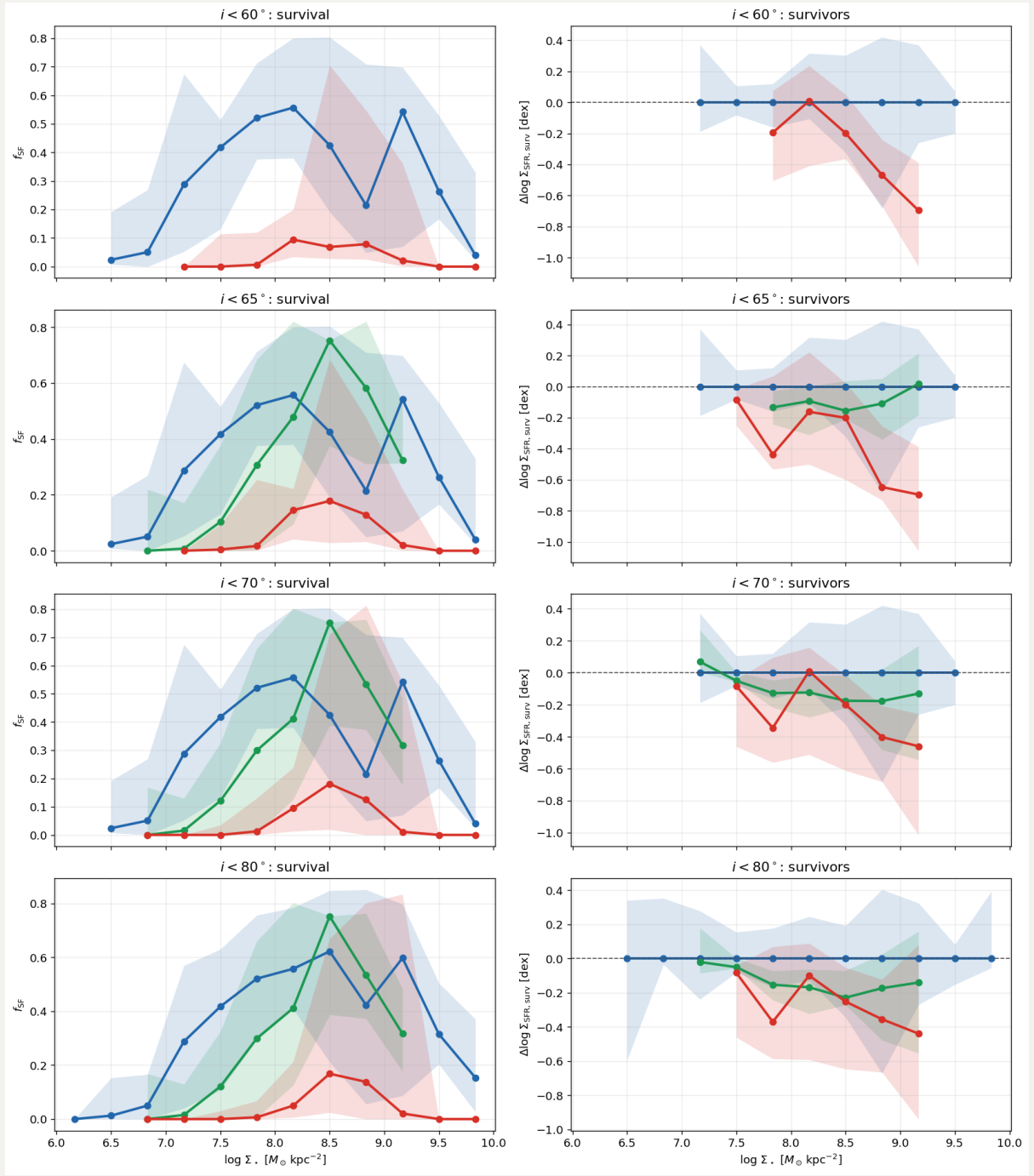


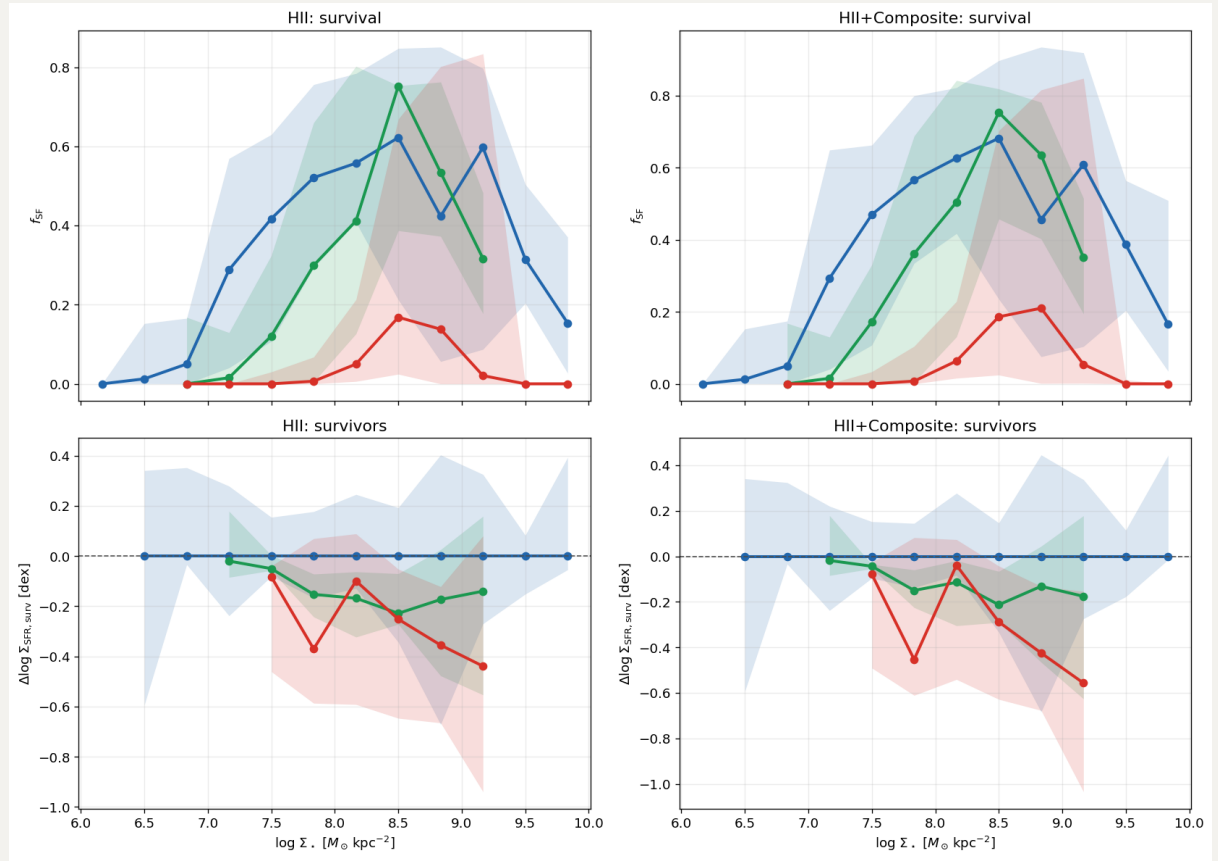
And in surviving SF intensity versus radius, median offsets fluctuate in all 3 stages around zero within R_e , and it is hard to tell if surviving SF regions in post-peak truly become less SF outside R_e .



4. Robustness test

Here I also show the $f_{\text{SF}} - \Sigma_*$ trend under different inclination selections and star forming selections.





5. Individual galaxy $f_{\text{SF}} - \Sigma_*$ trend

Finally, I show the $f_{\text{SF}} - \Sigma_*$ trend for each galaxy.

